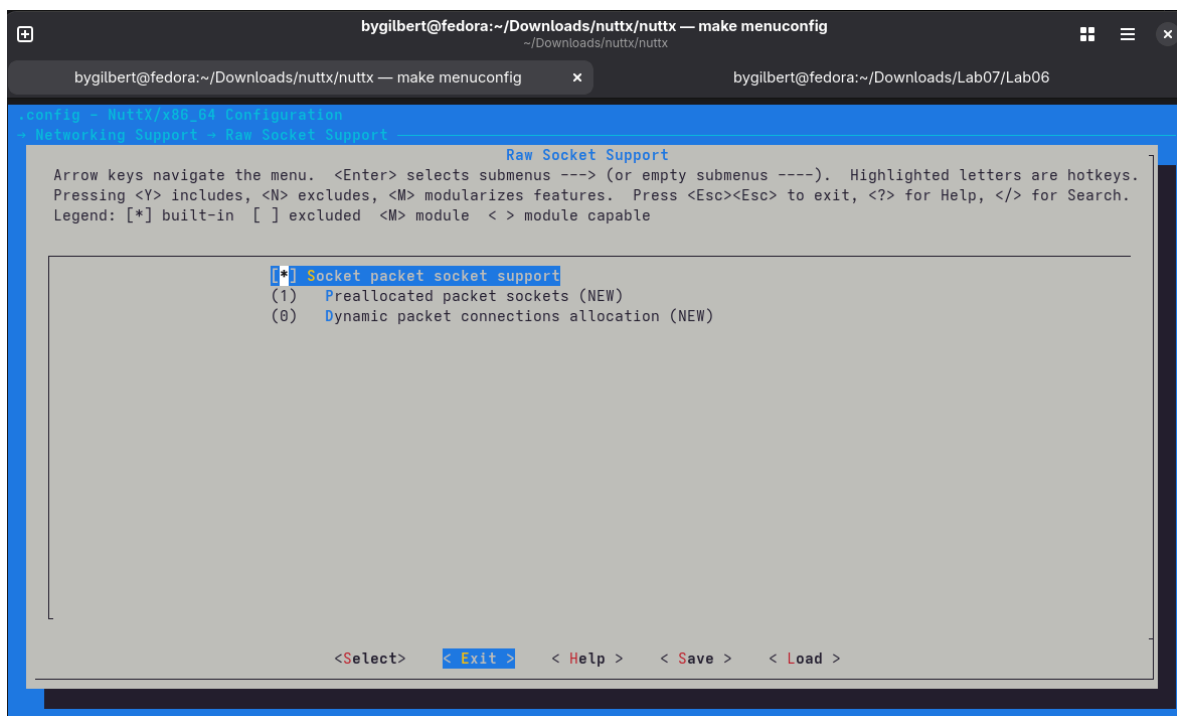


1. **Enable Essential RTOS Features (Dependencies):** Network drivers cannot perform long processes without IST so the work queue delegates this long running task to a Kernel Thread. SO the activation of these functions ensures that the system remains responsive and meets real time constraints

2. **Networking support:** it was activated the link layer where devices like Wi-Fi operate. Late driver initialization ensures that wlan0 loads after the kernel is stable, preventing startup conflicts. System Type and ESP32 Peripheral were activated so the necessary hardware configurations were enable for usage

3. **Connectivity and Utility Configurations:** these options configured how the device obtains an IP address (DHCP) and how the target Wi-Fi network was defined (WAPI configuration).

- The DHCP (Dynamic Host Configuration Protocol), used for automatic IP address assignment, relies on UDP Broadcast messages. These options enable the required message reception capability .
- The DNS support allowed the system to translate human readable names
- ICMP necessary for the ping command to work properly



```
bygilbert@fedora:~/Downloads/nuttx/nuttx — make menuconfig
~/Downloads/nuttx/nuttx

bygilbert@fedora:~/Downloads/nuttx/nuttx — make menuconfig
bygilbert@fedora:~/Downloads/Lab07/Lab06

config - NuttX/x86_64 Configuration
Networking Support - Internet Protocol Selection
Internet Protocol Selection
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module < > module capable

[*] IPv4
[ ] IPv6
[ ] Enable L2 forwarding
(1) Dynamic forwarding structures allocation
[ ] Enable IP packet filter (firewall)
[*] IP fragmentation support

<Select> < Exit > < Help > < Save > < Load >
```

```
bygilbert@fedora:~/Downloads/nuttx/nuttx$ grep "EMAC" .config
# CONFIG_ESP32_EMAC is not set
```

```
bygilbert@fedora:~/Downloads/nuttx/nuttx$ nano .config
bygilbert@fedora:~/Downloads/nuttx/nuttx$ grep "EMAC" .config
CONFIG_ESP32_EMAC=y
```

```
Link layer support
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module < > module capable

[*] Ethernet support
[ ] Cellular Link support
[*] Local loopback
(0) Loopback packet buffer size (NEW)
[*] MMIO modem support ----
[ ] SLIP support ----
[ ] TUN Virtual Network Device support ----
[*] Late driver initialization

<Select> < Exit > < Help > < Save > < Load >
```

```
bygilbert@fedora:~/Downloads/nuttx/nuttx — make menuconfig
config - Nuttx/x86_64 Configuration
File Systems
File Systems
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module <> module capable

↑(-)
[ ] ROMFS file system
[*] CROMFS file system
[ ] TMPFS file system
[ ] SMART file system
[ ] BINFS File System
[*] PROCFS File System
[ ] Run-time registration
    Exclude individual procs entries --->
-- SPIFFS File System
    *** In-memory Cache Options ***
(8192) Size of the cache
(4) Cache Hit Score
    *** Garbage Collection (GC) Options ***
(5) Max GC runs
(5) Deleted pages weight
(-1) Used pages weight
(50) Used pages weight
↓(+)

<Select> < Exit > < Help > < Save > < Load >
```

```
[ ] File mapping emulation
[*] Anonymous mapping emulation
    Partition Table --->
[ ] FAT file system
[ ] NFS client file system
[ ] NXFFS file system
↓(+)

<Select> < Exit > < Help > < Save >
```

```
config - Nuttx/x86_64 Configuration
Application Configuration -> Network Utilities
Network Utilities
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module <> module capable

↑(-)
(https://github.com/DaveGamble/cJSON/archive) URL where cJSON library can be downloaded
(1.7.12) Version number
[ ] Enable cJSON test
[ ] Cmux tool
[ ] CODEC Library
[ ] Connected Home over IP (CHIP)
-- DHCP client
(nuttx) DHCP client host name
(3000) Number of receive timeout in millisecond
(3) Number of retries for dhcp client request
(0x0000) Flags of Bootstrap
(3) DHCP Release send retries
(10) DHCP Release transmission delay (milliseconds)
[*] Ensure DHCP Release message transmission
[ ] Clear IP address after DHCP release
[*] DHCP server
[ ] Network Discovery Utility
↓(+)

<Select> < Exit > < Help > < Save > < Load >
```

```
[*] Ethernet support (NEW)
[ ] Cellular Link support (NEW)
[*] Local loopback
(0) Loopback packet buffer size (NEW)
[ ] MBIM modem support (NEW) ----
[ ] SLIP support (NEW) ----
[*] Late driver initialization
```

```
[*] Socket packet socket support
(1) Preallocated packet sockets (NEW)
(0) Dynamic packet connections allocation (NEW)
```

4. What was the workflow of activation

4.1 Wi-Fi Networking

- Network Support
 - Networking Support
 - Link Layer support
 - Late driver initialization
- System Type
 - ESP32 Peripheral Selection
 - Wireless
- Device Drivers

- Wireless Device Support
 - IEEE 802.11 Device Support
- Network Device Operation
 - Enable Wireless ioctl ()

4.2 Application and DHCP

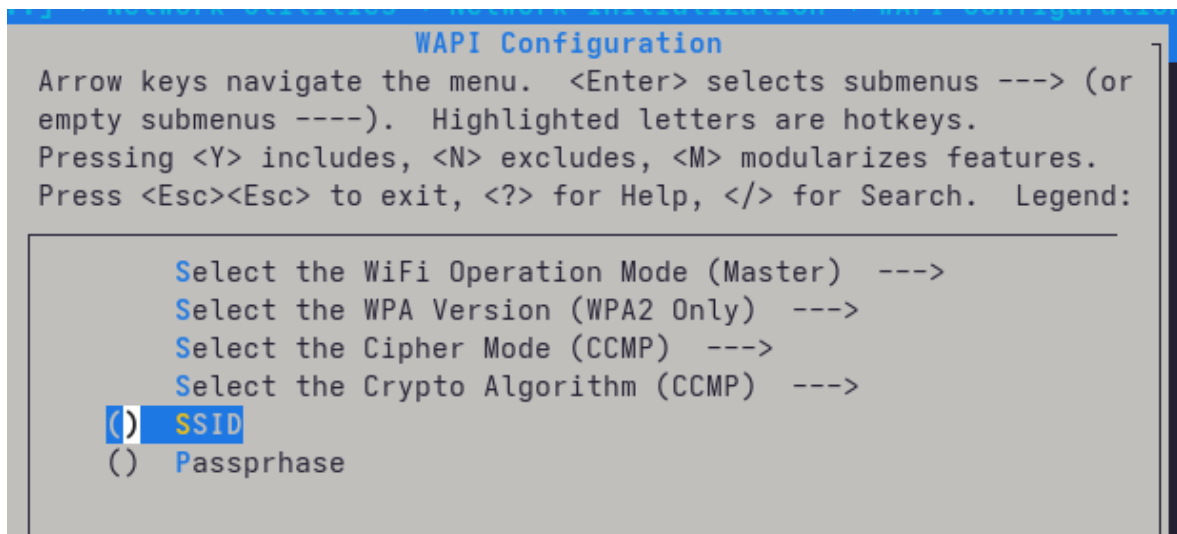
- Networking Support
 - UDP Networking
 - UDP broadcast RX support
- Application
 - Network Utilities
 - DHCP Client
 - Network Initialization
 - IP Address Configuration
 - Use DHCP to get IP address
 - WAPI Configuration (SSID/Passphrase)
- System Libraries and NSH Add-Ons
 - DHCP Address Renewal
- Wireless Libraries and NSH Add Ons
 - IEEE 802.11 Configuration Library

4.3 ICMP

- Networking Support
 - ICMP Networking Support

4.4 Debugging

- Buils Setup
 - Enable Error Output
 - Enable Debug Features
 - Network Debug Features
 - Wireless Debug Features



5. Root Cause Analysis: failed to initialize wi-fi station, wlan subsystem=-12: The error code -12 corresponds to the standard POSIX error ENOMEM (No memory), meaning the system failed to allocate the necessary memory to start the Wi-Fi station.

```
rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3ffb7dc0,len:11992
load:0x40080000,len:104684
1150 mmu set 00010000, pos 00010000
entry 0x400857b4
*** Booting NuttX ***
dram: lma 0x00001020 vma 0x3ffb7dc0 len 0x2ed8 (11992)
iram: lma 0x00003f00 vma 0x40080000 len 0x198ec (104684)
spadd: lma 0x0001d7f8 vma 0x00000000 len 0x2800 (10240)
simap: lma 0x00020000 vma 0x400f0000 len 0x77008 (487432)
spadd: lma 0x00097010 vma 0x00000000 len 0x8fe8 (36840)
sdmap: lma 0x000a0000 vma 0x3f410000 len 0x160a4 (90276)
total segments stored 6
ERROR: Failed to initialize Wi-Fi station
ERROR: Failed to initialize wlan subsystem=-12

NuttShell (NSH) NuttX-12.11.0
nsh>
```

```
total segments stored 6
ERROR: Failed to initialize Wi-Fi station
ERROR: Failed to initialize wlan subsystem=-12

NuttShell (NSH) NuttX-12.11.0
nsh> help
help usage:  help [-v] [<cmd>]

      .      cp      false    pkill      ps      true
      [      cmp      fdinfo    ls          pwd      truncate
      ?      dirname free      lsmod      rm        uname
alias      df      help      mkdir      rmdir    umount
unalias    dmesg   hexdump   mkrd      rmmod    unset
arp        echo    ifconfig  mount     set       uptime
basename   env      ifdown    mv         sleep     usleep
break      exec    ifup      nslookup  source    watch
cat        exit    insmod    pidof     test      xd
cd         expr    kill      printf    time      wait

Builtin Apps:
      dd      nsh      renew    sh      wapi
nsh> █
```

Builtin Apps:

dd nsh renew sh wapi

nsh> ls

/:

dev/

proc/

nsh> ifconfig

lo Link encap:Local Loopback at RUNNING mtu 1518

inet addr:127.0.0.1 DRaddr:127.0.0.1 Mask:255.0.0.0

nsh> █